

MARY COTRUPI

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Research interests: quantum software, quantum error correction & decoding, NISQ algorithms, high-performance computing

EDUCATION

Rensselaer Polytechnic Institute (RPI)

Troy, NY

B.S. & M.S. in Computer Science; Minors: Quantum Computing, IT & Web Science

B.S. May 2027 | M.S. Dec. 2027

Undergraduate GPA: 3.8/4.0 | Dean's List | RPI Founders Award of Excellence (top 1%)

Quantum coursework: Quantum Computing; Quantum Computing & Quantum Information; Quantum Information Processing; Quantum Computer Organization; Quantum Mechanics

CS/math coursework: Algorithms; Distributed & Parallel Computing; Linear Algebra; Machine Learning from Data; Discrete Mathematics & Probability; Operating Systems; Computer Architecture & Organization

The Chinese University of Hong Kong (CUHK) – Computer Science Study Abroad, Hong Kong

Spring 2026

TECHNICAL SKILLS

Programming & scientific computing: Python, C++, CUDA, CMake, NumPy, SciPy, JAX, Git/GitHub, Linux

HPC & performance: Slurm, MPI, OpenMP, Raft, NVIDIA Nsight, multi-GPU computing

Quantum computing: Qiskit, PennyLane, Qualtran, Stim, Aer, PyMatching, IBM Quantum, resource estimation

RESEARCH EXPERIENCE

Future of Computing Summer Intern (FCSI) | National Security Agency

Summer 2026

Towards Efficient Low-Weight Decoding: A Blocked Fast Walsh-Hadamard Transform

- Co-developed the blocked fast Walsh-Hadamard transform (BFWHT), an exact set-covering decoder for the low-weight binary CVP; at $n = 64$, $m = 2,097,152$, $t = 8$, reduced estimated operations from 9.28 quadrillion to 12.7 trillion ($732\times$) versus optimized Gray-code enumeration.
- Co-developed and benchmarked CUDA/GPU implementations on A100-class hardware, achieving a $43.6\times$ measured speedup at $n = 48$, $t = 8$ while evaluating covering designs, pruning, and memory/runtime tradeoffs.
- Built the end-to-end Decoded Quantum Interferometry (DQI) research codebase with interchangeable brute-force, full-FWHT, and BFWHT decoders and reversible resource estimation; reduced estimated logical qubits by $688\times$ on a 64×64 , $t = 8$ benchmark and executed small circuits on IBM's 156-qubit Kingston system.

Quantum Machine Learning Researcher | RPI School of Science

Summer 2025

Quantum machine learning kernels for network intrusion detection: a comparative NISQ application

- Designed a Qiskit quantum-kernel/SVM pipeline for network intrusion detection; achieved 85%+ accuracy and 6% simulator-to-hardware deviation on RPI's 127-qubit IBM Quantum System One versus classical RBF SVMs.
- Built the reproducible [QML-NIDS](#) research codebase for data preparation, hardware execution, benchmarking, and analysis; first-authored and presented the resulting AAAI QIML proceedings paper.

Cybersecurity Research Assistant | Rensselaer Cybersecurity Collaboratory

Aug 2024 – May 2025

- Developed augmented-reality CTF challenges in steganography, DNS poisoning, and MFA bypass; coauthored a 2026 CISSE article and ASIA proceedings on generative-AI use and cybersecurity pedagogy, presenting related work at BSides St. Pete.

SELECTED PUBLICATIONS & PRESENTATIONS

- Cotrupi, M. L.**, Gutti, M., Ralph, B., & Sandberg, K. (2026). *Towards Efficient Low-Weight Decoding: A Blocked Fast Walsh-Hadamard Transform*. FCSI 2026 final research report.
- Cotrupi, M. L.**, & Callahan, B. R. (2025). *Towards Practical Quantum Kernels for Network Intrusion Detection*. *Proceedings of the AAAI Symposium Series*, 7(1), 339–342. doi:10.1609/aaais.v7i1.36903
- Sugerman, S., Joseph, S., Colognato, Q., **Cotrupi, M. L.**, et al. (2026). *A Case Study for Combating Student Overuse of Generative Artificial Intelligence in Cybersecurity Educational Activities Using Augmented Reality Capture-the-Flag Development*. *Journal of the Colloquium for Information Systems Security Education*, 13(1), Article 11. doi:10.53735/cisse.v13i1.216
- Callahan, B. R., Joseph, S., Sugerman, S., Colognato, Q., **Cotrupi, M. L.**, et al. (2025). *RC3TF, an Augmented Reality CTF: A Case Study in Improving Cybersecurity Pedagogy for the Undergraduate Research Laboratory*. *Proceedings of the 20th Annual Symposium on Information Assurance (ASIA)*. doi:10.2139/SSRN.6550178
- BSides St. Pete (2024)** – Presented *Generative AI for Cybersecurity Awareness Training: Skills, Strategies, and Accountabilities*.

LEADERSHIP, ACTIVITIES & HONORS

- Cyber Sentinel Challenge**, U.S. Department of Defense – Top 5% among 2,100+ selected competitors (2025).
- Admin & Student Leader**, Rensselaer Cybersecurity Collaboratory – Led technical CTF sessions in reverse engineering, insider threats, and cyber forensics (2023–2025).
- Principal Second Violinist**, Rensselaer Orchestra – Section leader or assistant leader for approximately 15 violinists (2023–Present).
- RPI Quantum Computing Club** – Incoming external relations chair.
- Honors & professional societies** – RPI Upsilon Pi Epsilon (CS), RPI Gamma Nu Eta (ITWS), ISC2 Capital District NY.